Chapter 27 Problem Set

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1a. If $a_n = 1 - \frac{1}{n^3}$, then a_n converges to 1, and it is therefore Cauchy. To prove this, we need to show that for any $\epsilon > 0$, there exists $N > 0$ such that if $n > N$ then $|a_n - 1| < \epsilon$. Since $a_n = 1 - \frac{1}{n^3}$, $|a_n - 1| = \frac{1}{n^3}$. If $\frac{1}{n^3} < \epsilon$, then $n > \sqrt[3]{\epsilon}$. So setting $N = \sqrt[3]{\epsilon}$ works.

1b. If $a_n = \frac{2n^2}{n^2+1}$, then a_n converges to 2, and a_n is Cauchy. We know that $|a_n - 2| = |\frac{2n^2}{n^2+1} - 2| = |\frac{-2}{n^2+1}| = \frac{2}{n^2+1}$. So if $\frac{2}{n^2+1} < \epsilon$ then $n > \sqrt{2\epsilon - 1}$. Setting $N = \sqrt{2\epsilon - 1}$ guarantees that $|a_n - 2| < \epsilon$.

1c. If $a_n = \frac{n^2}{n+6}$, then it is not convergent or Cauchy. Given a number a, there exists n such that $a_n > a$. To see this, consider the inequality $\frac{n^2}{n+6} > a$. Then $n^2 > an + 6a$, or $n^2 - an - 6a > 0$. Since $n^2 - an - 6a$ represents an upward-opening parabola, there are values of n such that $n^2 - an - 6a > 0$. This means a_n grows without bound, so it cannot be Cauchy.

2. If $a_n = a$, then it converges to a. For all n and $\epsilon > 0$, $|a_n - a| = 0 < \epsilon$.

3. An example is the sequence $a_n = (-1)^n + 1$.

4. Since a_n is Cauchy, for all $m, n > N$, $|a_m - a_n| < \epsilon$ where $\epsilon > 0$. Since infinitely many values of n give $a_n = 0$, infinitely many of those values of n are greater than N. By the definition of a Cauchy sequence, we can find m greater than N such that $|a_m - 0| < \epsilon$ for all $\epsilon > 0$. This means a_n converges to 0.

5. Consider the sequence $a_n = \frac{1}{n}(-1)^n$. This converges to 0 because the absolute value is just $\frac{1}{n}$, which we know converges to 0. But when we take the floor, the new sequence switches between -1 and 0 at every n, so it is not convergent.

6. Let's say a_n converges to a. Then $|a_n - a| < \epsilon$ for all n greater than some $N > 0$ and $\epsilon > 0$. Since $n_1, n_2, n_3, \ldots$ is infinite, there is an infinite number of them greater than N. We know that $|a_n - a| < \epsilon$ for all those values, so the subsequence converges too.

7. Consider the sequence $a_n = (-1)^n$. The subsequence consisting of all even indices n converges to 1 because they are all within 0 of 1, but a_n itself doesn't converge.

8. A counterexample is the sequence $a_n = (-1)^n$. For every k, either $a_n = 1$ or -1. Then a subsequence can consist of all the other values of n that give $a_n = a_k$ (there are an infinite number of them). Since all terms of the subsequence are equal, it converges but a_n doesn't.

9. Assume for contradiction that a_n does not converge to r. Then there must exist an $\epsilon > 0$ such that for every $N \in \mathbb{N}$, there is some $k > N$ with $|a_k - r| \geq \epsilon$. But each such k is part of a subsequence converging to r, so within that subsequence, a_k will tend to r, which contradicts the existence of ϵ working for all N.